

Narratron: Collaborative Writing and Shadow-playing of Children Stories with Large Language Models

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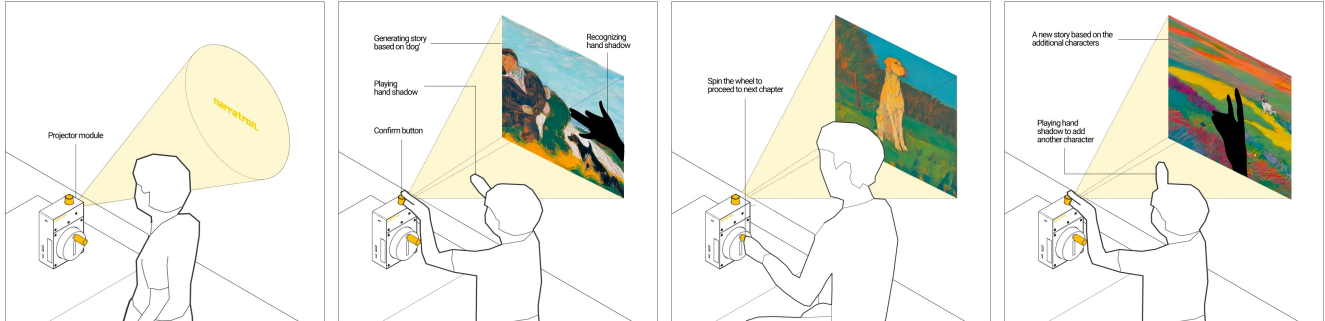


Figure 1: The demonstration of Narratron: (A) the projector module casts a light on the wall, (B) as the user poses the first shadow of a dog and confirms this input, the projector starts to cast a backdrop image for the Claude-2-generated story of the dog “Max”, and the speaker starts to narrate that story, (C) as the user spins the wheel, the story develops and the backdrop image changes according to the plot, (D) when the user decides to add a character by posing another shadow, the system adds a bunny named “Benny” into the plot, and the projected image as well as the narrated text all respond to that new story.

ABSTRACT

Shadow puppetry or shadow play, allows bodily participation into the process of linguistic storytelling, while the potential of multi-modal interaction through shadow plays in existing large-language-model-based creative tools has not been fully discovered. We propose Narratron, a generative story-making tool that co-creates and co-performs children stories from shadow using Claude 2 model. To achieve Narratron, our system is designed to recognize hand gestural inputs as main character and to develop story plot in accordance with character change. Through our system, we seek to stimulate creativity in shadow play storytelling and to facilitate a multi-modal human-AI collaboration.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Displays and imagers**; **Natural language interfaces**.

*Both authors contributed equally to this project.

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KEYWORDS

large language models, shadow play, storytelling, human-AI collaboration

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1 INTRODUCTION

Shadow play as a transcultural way of storytelling has been practiced and inherited in generations. It not only expresses human imagination through bodily interactions, but stimulates creativity and boosts early-childhood education[4]. However, while the literary capability of large language models has advanced rapidly in recent years for educational applications, it still communicates through textual I/O. This interaction paradigm does not suffice to comprehensively contribute to learning process for young children whose literacy is essentially multi-modal, as a combination of oral narration, visual images, and body engagement [8].

The participation of large language model in shadow play storytelling process can also push the boundary of human-AI interaction by expanding the story co-creation experience to multi-sensory immersion[2]. To achieve this, our system integrates three major systems: 1. hand gesture recognition through custom-trained image classifier to create animal character, 2. 5-sentence short children story generated by pre-trained Claude 2 model with ability to add

a second character, 3. visual and audio outputs related to the story through diffusion model and speech synthesizer. To allow body engagement and provide physical affordance of "making stories" for young children, we decide to design our hardware as a toy-like device with a film projector knob and a camera shutter instead of relying on web or mobile devices (Figure 2 in Appendix A).

2 CHILDREN STORY CREATION

We now introduce our approach to plan-based story writing that implements few-shot learning and instruction-following methods. The pre-trained language model is prompted with a few examples, given a set of instructions to continue the story with the addition of a second character, and tasked to generate the story. The instances from examples include 1. the species and a name of main animal character which is performed by hand shadow, 2. the three-act structure as the plot outline, 3. the optional continuation prompt that includes instructions for second character addition into the plot, if a second character is performed. For the example dataset, we fetched 10 children stories with approximately 1000 words from open source websites and condensed them into 5 sentences while generating all instances required by the examples by using Claude 2 ($temp = 0.8$), and only one of ten is shown in Appendix C for brevity.

2.1 Three-act Structure

Many previous studies have suggested that three-act structure is an effective planning method for LLM-based story generation [7]. We exemplified this structure by following this format: Setup, Confrontation, and Resolution. The prompting of this structure is 1. What is the setup of this story? 2. What is the conflict? 3. What is the resolution? The result given by this structure accomplished the desired coherence and attractiveness of the story we anticipated.

2.2 Continuation and Second Character Addition

The addition of a second character as a crucial part of plot development poses two challenges to our system. First, an unpredictable input of new character often weakens logical progression, such as repeated uninteresting storyline between two animal characters, or even illogical development of the story. For instance, we discovered that the plots developed from a first character of a dog to the addition of a second character of either a bunny or a lion are almost the same, which fails to sensemaking since the relationship in latter pair should be way less friendly than the first. Secondly, as stories with multiple characters serve as a primary media for shaping children's moral values in social interaction[6], we discovered that a plot change brought by a second character often leads to only one happy ending with vague moral orientation or no moral lessons.

To address these concerns, we developed a set of instructions for our examples that is added to the three-act structure when the user inputs a second character.

2.2.1 Relationship to Main Character. To improve the coherence and logic after the addition of a second character, we instructed the model to: 1. Define the relationship between <main character> and <second character> before the resolution (friendly, or neutral, or

adversarial), 2. Define the relationship between <main character> and <second character> after the resolution (close connection, or respected competitor, or winner versus loser).

2.2.2 Moral Value Establishment. We provided an instruction to define moral values in order to enhance relational dynamics in multi-character settings. The model is prompted as following when a second character joins the story: Define the moral value(s) of the plot (choose one or more from care, fairness, bravery, and loyalty).

3 INTERFACE

We now introduce how the core story generation module integrates with three other AI-content-generation pipelines to achieve the multi-modal experience (Figure 3 in Appendix A). First, we trained an image classifier that can recognize 10 hand-posed animals by using Google Teachable Machine[5] interface and integrated it to our system through Tensorflow.js API. Secondly, we trained Stable Diffusion 2.0 model by using Low-Rank Adaption method through Replicate (a cloud-based API[1]) for children-friendly image generation. Thirdly, we connected the textual output to a react-speech-kit for oral narration.

4 LIMITATIONS AND FUTURE STEPS

After few-shot learning process the story results still occasionally lack excitement and are filled with incoherence. We also realized certain ethical issues relating to children's exposure to existing LLMs which could possibly cause trustworthiness and poor content moderation problems [3] (see Appendix B). To respond that, our next step is to: 1) evaluate with educators for trustworthiness check and with young users for usability test, 2) explore furthermore to improve the coherence and after all, the "interestingness" of the storyline.

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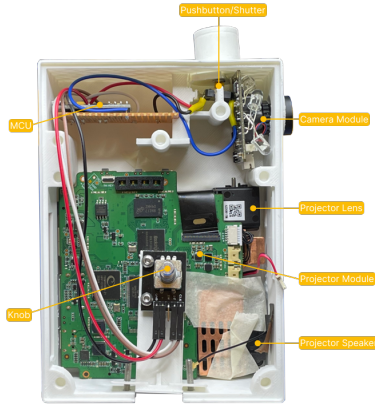


Figure 2: Toy-like Device as Tangible Interface with Multiple Input and Output Components

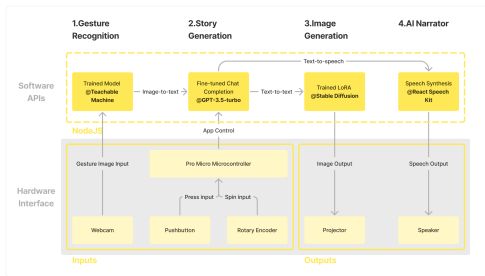


Figure 3: Information Architecture and Shadow Play Co-writing and Co-performing Pipelines

A HARDWARE AND INFORMATION ARCHITECTURE

B ETHICAL AND BROADER IMPACT STATEMENT

B.1 Applications: Impact on ML Applications

Narratron serves as a unique example of how machine learning, specifically natural language processing, can be integrated with traditional arts like shadow puppetry. The project demonstrates the potential for AI to be used in creative storytelling contexts, thereby diversifying its application beyond conventional domains. It might catalyze similar multimodal, human-AI collaborative projects in other artistic or educational disciplines, such as music, visual arts, or pedagogical tools. The utilization of machine learning for real-time interpretation of hand poses to generate textual content could also find applications in gesture-based human-computer interactions, virtual gaming, and assistive technologies for individuals with speech or motor impairments.

B.2 Implications: Societal Implications of These Applications

B.2.1 Educational Efficacy. By amalgamating kinesthetic and cognitive learning activities, Narratron could serve as a valuable pedagogical tool. This could be particularly impactful in early childhood education, enhancing engagement and learning outcomes.

B.2.2 Cultural Sensitivity. There is a risk that the algorithm could inadvertently reinforce existing sociocultural stereotypes, particularly if the data used for training the model is not carefully curated. This could have societal implications, such as contributing to existing prejudices and biases. The algorithm may assign specific traits and roles to animal characters, which may be inadvertently derived from predetermined, culturally-biased categorizations.

B.2.3 Data Privacy. With the increasing integration of AI in personal and educational spaces, concerns about data privacy and information security become paramount. If not managed appropriately, this could undermine trust in such human-AI collaborative systems, especially if the system stores or processes images or personal information.

B.3 Initiatives: Research or Other Initiatives to Improve Societal Outcomes

B.3.1 Algorithmic Fairness and Transparency. Future research could focus on developing more transparent and interpretable algorithms to ensure that the storytelling process is free from bias. Collaboration with experts in ethics and social sciences could help to scrutinize the societal impact of the technology more closely.

B.3.2 Data Privacy Protocols. Establishing robust data encryption and anonymity measures, in conjunction with research into privacy-preserving machine learning models, could significantly mitigate concerns about data privacy. This kind of study is especially crucial for such human-AI collaboration application that requires input and output from both side.

B.3.3 Pedagogical Studies. Longitudinal studies could be conducted to examine the efficacy of Narratron as a learning tool, which would provide data-driven insights into its potential educational benefits or drawbacks. Research could also be conducted to explore the potential of combining other kinds of mediums, such as immersive projection, music creation, art installation to bring this educational value to a broader user group. By conscientiously considering these factors and implementing research initiatives aimed at mitigating negative impacts, Narratron aims to be a socially responsible addition to the intersection of AI and human creativity.

C PROMPT ENGINEERING

This appendix shows our preparation of raw story data fetched from online resource and translated to 5-sentence short story and 5 image prompts (Table 1) and few-shot learning process (Table 2) with three-act structure (only showing one of ten total examples for brevity) and optional provided instruction if a second character is added (Table 3).

Prompt Type	Role	Prompts
Raw Story Data	Human	Rewrite the story to a 5-sentence short version with 5 scenes and generate 5 image prompts according to each scene for Stable Diffusion: Mummy Duck lived on a farm. In her nest, she had five little eggs and one big egg. One day, the five little eggs started to crack. Tap, tap, tap! Five pretty, yellow baby ducklings came out. Then the big egg started to crack. Bang, bang, bang! One big, ugly duckling came out. ‘That’s strange,’ thought Mummy Duck. Nobody wanted to play with him. ‘Go away,’ said his brothers and sisters. ‘You’re ugly!’ The ugly duckling was sad. So he went to find some new friends. ‘Go away!’ said the pig. ‘Go away!’ said the sheep. ‘Go away!’ said the cow. ‘Go away!’ said the horse. No one wanted to be his friend. It started to get cold. It started to snow! The ugly duckling found an empty barn and lived there. He was cold, sad and alone. Then spring came. The ugly duckling left the barn and went back to the pond. He was very thirsty and put his beak into the water. He saw a beautiful, white bird! ‘Wow!’ he said. ‘Who’s that?’ ‘It’s you,’ said another beautiful, white bird. ‘Me? But I’m an ugly duckling.’ ‘Not any more. You’re a beautiful swan, like me. Do you want to be my friend?’ ‘Yes,’ he smiled. All the other animals watched as the two swans flew away, friends forever.
Rewritten 5-sentence Story	Assistant	Here is a 5-sentence short version of the story with image prompts: Once upon a time, there was an ugly duckling who was born into a family of ducks. He was constantly ridiculed and bullied for his appearance, so he ran away from home and found himself alone in the world. Over time, the duckling grew into a beautiful swan, and as he swam in a pond, the other animals marveled at his grace and beauty. They soon realized that the once-ugly duckling had transformed into a majestic swan. The swan was overjoyed to find that he belonged somewhere, and he lived the rest of his life surrounded by friends who loved him for who he truly was.
Rewritten 5-line Image Prompt	Assistant	1. A small pond with lush greenery and a bright blue sky. 2. A fuzzy, out of focus view of some tall grass and bushes, with glinting beams of morning sunlight. 3. A tranquil lake surrounded by lush green trees, with a beautiful swan swimming elegantly in the water. 4. A sun-dappled garden scene with soft touches of bright foliage. 5. A magical meadow filled with colorful wildflowers and butterflies.

Table 1: Preparing Raw Story Data for Few-shot Learning (Light Cyan indicates results generated by Claude 2 model)

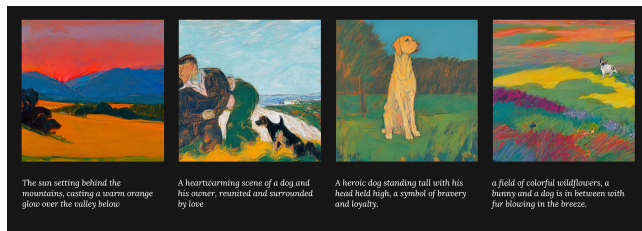


Figure 4: Image prompts and Generated Images

D IMAGE GENERATION

The image prompts generated from the story and generated images by LoRA trained diffusion model (Figure 4):

1. A small pond with lush greenery and a bright blue sky.
2. A fuzzy, out of focus view of some tall grass and bushes, with glinting beams of morning sunlight.
3. A tranquil lake surrounded by lush green trees, with a beautiful swan swimming elegantly in the water.
4. A sun-dappled garden scene with soft touches of bright foliage.
5. A magical meadow filled with colorful wildflowers and butterflies.

Few-shot Learning Example		
Human	Three-act Structure Sub-task Prompt	Generate a 5-sentence children story based on a { duck } SOLELY. Before writing, answer these questions as an outline briefly. Only respond with the outline, one sentence per line, nothing else: 1. What is the setup of this story? 2. What is the conflict? 3. What is the resolution?
Assistant	Three-act Structure Sub-task Result	Here are the answers as an outline before writing: 1. What is the setup of this story? A duck named Ga was born with ugly appearance. 2. What is the conflict? He is constantly ridiculed and bullied so one day he left home alone. 3. What is the resolution? He turned into a beautiful swan when came back with friends who truly love him.
Human	Three-act Structure Sub-task Validation	Great, now write a 5-sentence story based on the outline.
Assistant	Three-act Structure Story Result	Here is the 5-sentence short story written based on the outline: Once upon a time, there was an ugly duckling named Ga who was born into a family of duck. He was constantly ridiculed and bullied for his appearance, so he ran away from home and found himself alone in the world. Over time, the duckling grew into a beautiful swan, and as he swam in a pond, the other animals marveled at his grace and beauty. They soon realized that the once-ugly duckling had transformed into a majestic swan. The swan was overjoyed to find that he belonged somewhere, and he lived the rest of his life surrounded by friends who loved him for who he truly was.
Human	Image Prompt Generation	Great, now generate 5 image prompts based on the landscape scene from each sentence of the story.
Assistant	Image Prompt Result	Here are 5 image prompts generated based on the landscape scene from each sentence of the story: 1. A small pond with lush greenery and a bright blue sky. 2. A fuzzy, out of focus view of some tall grass and bushes, with glinting beams of morning sunlight. 3. A tranquil lake surrounded by lush green trees, with a beautiful swan swimming elegantly in the water. 4. A sun-dappled garden scene with soft touches of bright foliage. 5. A magical meadow filled with colorful wildflowers and butterflies.
Human	Story Generation	Generate a 5-sentence children story based on a { dog } SOLELY. Before writing, answer these questions as an outline briefly. Only respond with the outline, one sentence per line, nothing else: 1. What is the setup of this story? 2. What is the conflict? 3. What is the resolution?
Story Completion		
Assistant	Prefix	Here are the answers as an outline before writing:

Table 2: Few-shot Learning Example

Human	Instruction Prompt	Extend the generated story to 5 more sentences by adding the second character { bunny }. Follow the steps first and write nothing else: 1. Define the relationship between main character { dog } and second character { bunny } before the resolution (friendly, or neutral, or adversarial). 2. Define the relationship between main character { dog } and second character { bunny } after the resolution (close connection, or respected competitor, or winner versus loser). 3. Define the moral value(s) of the plot (choose one or more from care, fairness, bravery, and loyalty).
Assistant	Prefix	Here are the answers: 1. Define the relationship between main character { dog } and second character { bunny } before the resolution: 2. Define the relationship between main character { dog } and second character { bunny } after the resolution: 3. Define the moral value(s) of the plot:

Table 3: Optional Instruction (Only if a second character joins)